‘ecosystems’ and ‘communities’ consisting of many species. In the long term, I expect the ESS concept to revolutionize the science of ecology.

We can also apply it to a matter that was deferred from Chapter 3, arising from the analogy of oarsmen in a boat (representing genes in a body) needing a good team spirit. Genes are selected, not as ‘good’ in isolation, but as good at working against the background of the other genes in the gene pool. A good gene must be compatible with, and complementary to, the other genes with whom it has to share a long succession of bodies. A gene for plant-grinding teeth is a good gene in the gene pool of a herbivorous species, but a bad gene in the gene pool of a carnivorous species.

It is possible to imagine a compatible combination of genes as being selected together *as a unit*. In the case of the butterfly mimicry example of Chapter 3, this seems to be exactly what happened. But the power of the ESS concept is that it can now enable us to see how the same kind of result could be achieved by selection purely at the level of the independent gene. The genes do not have to be linked on the same chromosome.

The rowing analogy is really not up to explaining this idea. The nearest we can come to it is this. Suppose it is important in a really successful crew that the rowers should coordinate their activities by means of speech. Suppose further that, in the pool of oarsmen at the coach’s disposal, some speak only English and some speak only German. The English are not consistently better or worse rowers than the Germans. But because of the importance of communication, a mixed crew will tend to win fewer races than either a pure English crew or a pure German crew.

The coach does not realize this. All he does is shuffle his men around, giving credit points to individuals in winning boats, marking down individuals in losing boats. Now if the pool available to him just happens to be dominated by Englishmen it follows that any German who gets into a boat is likely to cause it to lose, because communications break down. Conversely, if the pool happened to be dominated by Germans, an Englishman would tend to cause any boat in which he found himself to lose. What will emerge as the overall best crew will be one of the two stable states—pure English or pure German, but not mixed. Superficially it looks as though the coach is selecting whole language groups *as units*. This is not what he is doing. He is selecting individual oarsmen for their